F005 v0.4.0 | Free-run & family audit

PRE-RUN SPECIFICATION | 19 October 2026 | Development only

The actual 9,822 x 1,500 measured robot dataset is stored on the user's local Windows workstation and has not been accessed by the authoring environment. Results of the v0.4.0 **real-data run are pending**. Synthetic unit tests and a full-size synthetic end-to-end smoke test have passed.

Known input from v0.3.0

Pre-existing locked ARX(2) parameters were fitted on the 3,500 candidate TRAIN series: A=1.9856661039519918, B=-0.9908696423084401, C=-1.118156917616826. Time increment dt=0.004 s. Earlier one-step validation error: 0.0156504 mm. This **does not** establish multi-second forecast validity.

Free-run equation and units

```
eta[n+1] = A eta[n] + B eta[n-1] + C (dt)^2 a[n]
```

Elevations eta are in m and the exogenous acceleration a in m/s^2; all coefficients are dimensionless. For each of three predeclared anchors, two observed heights seed the recursion. Thereafter the **predicted** heights are recycled, without teacher forcing. The full future input acceleration is supplied as known external forcing.

Experimental design (locked before local execution)

Variable	Predefined values
Validation	750 STANDARD_VALIDATION response trajectories
Forecast anchor	0.004, 1.5, 2.8 s
Forecast horizons	0.1, 0.5, 1.0, 2.0, 3.0 s
Comparators	Persistence; linear trend extrapolation
Metrics	RMSE in m and mm; max absolute height
Family matching	Only input accelerations, exact + 3 multi-resolution fingerprints

P5b - independent trajectory-family screening

Generate candidate matches from amplitude-normalized acceleration shapes (64 bins / 0.05, 32 bins / 0.1, 16 bins / 0.20). For candidates, require Pearson r >= 0.995 and normalized RMS difference <= 0.075 allowing small timing shifts. Exact binary duplicate inputs are audited separately. Oversized buckets and budget exhaustion remain explicit open issues. **No waveform screening can independently establish publisher-grounded trajectory-family IDs.**

Holdout protection and scope

TEST (750) and OOD (4,822) output/elevation rows are never indexed, hashed, plotted, or fitted. Their external input/acceleration rows may be read **only for data-leakage screening**. No operation opens the ~49.9 GB DATASET.zip file. The current split remains candidate-only / non-blind.

Qualification gates

G0: Pre-existing hashes, time vector, dimensions match; else BLOCK

G1: Waveform-family candidates computed, provenance remains REVIEW

G2: Multiple conditional free-run forecasts evaluated on VALIDATION only

G3: Persistence and extrapolation baselines compared at every horizon

G4: No TEST/OOD output use; pre-freeze requirements remain OPEN

G5: Euler/PKLSA/physical SST validation NOT RUN / UNRESOLVED

Synthetic qualification completed

16/16 small synthetic regressions PASS. One 9,822-row synthetic integration test PASS with deliberately NaN-poisoned TEST/OOD elevations and one inserted cross-partition duplicate acceleration. A four-page runtime PDF was generated and visually checked. None of these are empirical validation.

Execute and return

From your existing F005-v0.1.0 directory:

```
py -3 -m pip install numpy matplotlib
.\data\run_f005_v040.cmd
```

Return outputs/f005_free_run_family_v0.4.0/v040_report.json and V040_SUMMARY.md; optionally CSV/PDF. Keep TEST/OOD sealed until source-defined groups and physics predictions are frozen.

References and status

Robot-executed sloshing dataset, Zenodo doi:10.5281/zenodo.20039757; L. Ljung, *System Identification: Theory for the User*, 2nd ed., Prentice Hall (1999), ISBN 9780136566953. Source dataset identification and complete contributor list must be verified for a formal publication. No SST-specific physical claims are implied.